HDOC Day-51: Previous Greater Element

Aim

To write a C program that reads a variable-length integer array and finds the nearest greater element on the left of every array element using a function named **previous_greater()** and pointer-based array passing.

Problem Statement

Read the size and elements of a variable-length integer array, **arr[]**. Pass the array to **previous_greater()** using its address. For each element, search only the elements to its left and select the nearest element that is greater than the current element. Display **-1** when no greater element exists. Print the results in array order, separated by commas.

Algorithm

1. Read the size **n** of the array.
2. Read **n** integer elements into the variable-length array.
3. Call **previous_greater(arr, n)**. The array is passed through its address.
4. For each index **i**, start checking from index **i - 1** and move left.
5. Stop at the first element greater than **arr[i]**; this is the nearest previous greater element.
6. If no greater element is found, use **-1**.
7. Print all answers in order, separated by commas.

C Program

```
#include <stdio.h>

void previous_greater(int *arr, int n)
{
    int i, j, greater;

    printf("Previous greater elements: ");

    for (i = 0; i < n; i++)
    {
        greater = -1;

        for (j = i - 1; j >= 0; j--)
        {
            if (*(arr + j) > *(arr + i))
            {
                greater = *(arr + j);
                break;
            }
        }

        printf("%d", greater);
        if (i < n - 1)
            printf(", ");
    }

    printf("\n");
}

int main()
{
    int n, i;
```

```
    printf("Enter size of array: ");
    scanf("%d", &n);

    int arr[n];

    printf("Enter %d elements: ", n);
    for (i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    previous_greater(arr, n);

    return 0;
}
```

Test Cases

Test Case	Input Array	Expected Previous Greater Elements
1	10, 4, 2, 20, 40, 12, 30	-1, 10, 4, -1, -1, 40, 40
2	5, 5, 5, 5	-1, -1, -1, -1
3	9, 7, 8, 3, 6	-1, 9, 9, 8, 8

Sample Output 1

```
Enter size of array: 7
Enter 7 elements: 10 4 2 20 40 12 30
Previous greater elements: -1, 10, 4, -1, -1, 40, 40
```

Sample Output 2

```
Enter size of array: 5
Enter 5 elements: 9 7 8 3 6
Previous greater elements: -1, 9, 9, 8, 8
```

Result

The program successfully finds and displays the nearest previous greater element for every array element using pointers. If no greater element exists on the left, it displays -1.